

Predicting Eurex bond futures with *TLOB*

A deep-learning directional signal and a market-making
backtest across the euro govvie curve — Citi POC prep, week
of 26 Jun – 2 Jul 2026.

WHAT I WORKED ON THIS WEEK

Four questions, one model

- 1

Reframing the label for market making

Why the spread-gated target is unlearnable at 1s – and the fix.
- 2

TLOB vs Tushar's linear model

Head-to-head across the curve, and where the DNN actually wins.
- 3

Does more input help?

Order-book depth, cross-asset stacking, micro features.
- 4

A realistic market-making backtest

Queue-aware maker with adverse selection – does the signal pay?



SETUP

Data & model

- **Six euro govvie futures** — Bund, Bobl, Schatz, Buxl (DE), BTP (IT), OAT (FR).
- **Databento TBBO, top-of-book (L1)** — ~252 trading days, 2025-06 → 2026-05.
- **TLOB** — deep transformer over the raw LOB sequence.
- **3-class target** — up / flat / down, chronological 80/5/15 split, out-of-sample argmax.

Anchor instrument — Bund (FGBL)	
Tenor	10y German
Feed	TBBO / L1
Input width	8 features
OOS test rows (1s)	~340k
Baselines	random 33% · majority-flat 34%



THE KEY METHODOLOGICAL MOVE

A label a maker can actually learn

- **The problem** – the standard label gates a move on *spread + fee*. At 1s almost nothing clears ~2 ticks → ~97% "flat" → degenerate to learn.
- **The insight** – as a *maker* we never cross the spread, so drop the gate.
- **New target** – sign of the next-second smoothed-mid drift → a balanced ~33/33/33 up/flat/down.
- **Framework** – added label_threshold_ticks to tlob, freeing fee for the backtest.

Class balance @ 1s

Spread+fee gate	~97% flat ✕
sign(Δ smoothed-mid)	29 / 43 / 29 ✓
Result	learnable, balanced

The reframing is what makes every result that follows possible.



RESULT 1 – THE DNN CASE

TLOB beats the linear model by +12 pts at 1s

🕒 HORIZON: 1S – TIME-BASED

MODEL – BUND, 1S SIGN TARGET	ACCURACY	MACRO-F1	FLAT F1
TLOB (deep, raw L1 sequence)	0.611	0.609	0.669
Linear – Tushar’s full 40-feature L1 set	0.489	0.472	0.316
Δ (TLOB – linear)	+0.122	+0.137	+0.353

Identical target, split and ~340k OOS rows. The biggest gap is the **flat** class (+0.35 F1): the linear model barely detects "no-drift" periods (0.316) while TLOB reads them well (0.669). Imbalance alone – ~92% on the mechanical next-tick target – reaches only ~49% here.



1s time horizon vs 1 / 2 / 5-event tick horizons

→ BUND · TLOB · 3-CLASS PURE-SIGN · TBBO L1

HORIZON	ACC	MACRO-F1	UP F1	FLAT F1	DOWN F1	FLAT %
1 second — time-based	0.611	0.609	0.596	0.669	0.561	43%
1 event — tick-based	0.861	0.785	0.724	0.908	0.723	~82%
2 events — tick-based	0.743	0.708	0.666	0.802	0.656	—
5 events — tick-based	0.681	0.681	0.685	0.681	0.678	40%

Tick horizons post a higher headline — but read the flat column. At 1 event the target is ~82% "flat", so accuracy 0.861 / flat-F1 0.908 are inflated by the easy class; the honest read is the up/down F1 (~0.72). As the event horizon lengthens (1 → 5) the flat share falls toward balance (~82% → 40%) and the up/down F1 stays strong. **The 5-event target is the most balanced tick target and is genuinely more predictable than 1s (0.68 vs 0.61 macro-F1) — a 5-event smoothed drift persists more than a 1-second one.**



The same liquidity gradient, top to bottom

🕒 HORIZON: 1S – TIME-BASED

INSTRUMENT	TENOR	ACCURACY	MACRO-F1	UP F1	DOWN F1
Bund (FGBL)	10y DE	0.611	0.609	0.596	0.561
Bobl (FGBM)	5y DE	0.580	0.576	0.566	0.529
BTP (FBTP)	10y IT	0.571	0.521	0.602	0.631
Buxl (FGBX)	30y DE	0.561	0.546	0.586	0.616
OAT (FOAT)	10y FR	0.558	0.541	0.592	0.594
Schatz (FGBS)	2y DE	0.543	0.542	0.513	0.536

German core on top, Schatz weakest – the same ordering as Tushar’s report. BTP / Buxl / OAT have low flat-F1 but the sharpest *directional*F1: their mid rarely sits still at 1s, so accuracy is dragged down by a thin flat class.



Order-book depth *hurts*. Top-of-book is enough.

🕒 HORIZON: 1S – TIME-BASED · MATCHED 21-DAY WINDOW

0.557

TBBO – L1, 8 features

Top-of-book only, matched 21-day window.

More inputs on far less data is less data-efficient, and depth carries little marginal signal at 1s. The same story repeats at the next-tick horizon (0.747 with L10 vs 0.793 with L1). **Use TBBO / L1** – also the only feed available across all six instruments and the full year.

0.497

MBP-10 – L10, 44 features

Full depth: -6.0 pts, and it overfits fast.

RESULT 4 – THE OPPOSITE REGIME

At the next tick, TLOB only ties the linear

→ HORIZON: NEXT TICK – EVENT-BASED, BINARY

MODEL – BUND, NEXT-TICK SIGN (BINARY)	FEATURES	SIGN ACCURACY
Linear – imbalance only	1	0.7924
Linear – Tushar’s full L1 set	40	0.7926
TLOB (deep)	raw L1 seq	0.7932
TLOB – MBP-10 full depth	44	0.7472

Replicated Tushar’s exact next-tick label inside our framework. It is **near-mechanical**: top-of-book imbalance alone hits ~79.2%, the other 39 features add +0.02 pt, and TLOB matches – **no deep-model edge here**. A sequence model has nothing to add once the static imbalance signal fully determines the answer.



THE INSIGHT, IN ONE LINE

TLOB's edge over linear grows with the horizon

Next tick

mechanical · imbalance
solves it



1s smoothed

signal has decayed ·
sequence matters



A sequence model earns its keep only once the mechanical top-of-book imbalance signal decays. At the next tick the answer is already in the current book; at 1s, the past trajectory carries the information the linear model can't see.



RESULT 5 – FEATURE ENGINEERING

Micro features give a small, consistent lift

+1.0 pt

1s sign target

0.611 $\rightarrow$ 0.621 accuracy on Bund.

Ported the linear baseline's signals – notional-imbalance ratio, VWAP difference, EWMA signed-flow / OFI, micro-price premium – into the DNN input (add_micro_extra, 22 features total). A $\sim +0.7$ – 1.0 pt lift on both targets, matching the order of magnitude we expected.

+0.75_{pt}

5-tick sign target

0.681 $\rightarrow$ 0.689 accuracy on Bund.



A queue-aware market-making simulation



Earn the spread

Post one clip at best bid & ask. Maker fills pay **no fee**; only the close-out crosses (€0.40/side).



Trade-flow L1 queue

Join the back of the queue; real aggressor volume from the tape consumes it before our clip fills.



Adverse selection

If the market trades through our stale quote we're filled underwater – the core cost being measured.



Pull the toxic side

Predict up → pull the ask, keep the bid; mirror for down; quote both when flat.



Risk controls

Hard inventory cap ± 50 lots, flatten at the 17:00 Berlin close.



Baseline control

Same maker, signal **off** – quotes both sides always. Signal minus baseline = what the prediction adds.

RESULT 6B – DOES THE SIGNAL PAY?

The 1-tick TLOB maker is ~breakeven

BUND MAKER	NET P&L	SHARPE	FILLS	PER-FILL
TLOB – 1-tick, flat-aware	-14.8k	-0.80	~55k	-0.27
No-signal two-sided baseline	-124.6k	-5.6	~82k	—
Linear (2-class, no flat) as a maker	similar	-4.32	1,147	-11.29

With the fee fix + realistic queue, the signal cuts adverse selection and lifts P&L **+29–88%** over a blind maker. **A flat class is essential:** the 2-class next-tick signal has no "no-move" state, so the maker is forced one-sided, flips constantly, and **barely trades (1,147 fills) – non-viable despite 79% directional accuracy**. Still marginally negative; closing the gap needs queue priority / confidence-weighted skew.



RECAP

Six takeaways

1 • The reframing works

Dropping the spread+fee gate makes the 1s direction target learnable (vs 97%-flat).

2 • TLOB > linear at 1s

+12 pts accuracy on the identical target – the clear DNN case.

3 • Edge grows with horizon

A statistical tie at the mechanical next-tick; the DNN wins once the imbalance signal decays.

4 • L1 is enough

MBP-10 depth hurts and overfits at both horizons. Top-of-book carries the signal.

5 • Micro features add ~1 pt

Small but consistent lift on 1s and 5-tick targets.

6 • Maker ~breakeven

Signal cuts adverse selection; the flat class is what makes a viable maker.



Next: close the *breakeven gap*

A proper maker model — **queue priority + confidence-weighted quote skew** — where earning the spread should turn the economics positive.

Lead-lag & cross-asset detection covered in a separate deck.